

Project

Heart Disease Prediction Using Amazon SageMaker Canvas: A No-Code Approach

Test this project if you're a business analyst, AWS AI/ML beginner, certification candidate (AIF-C01 / MLA-C01), or career switcher looking to build and deploy a real ML model on AWS without writing any code.

Project Scenario:

Heart disease is the world's leading cause of death, and the cost of missing an at-risk patient is enormous both in lives and in hospital expenses. Most healthcare providers already sit on years of patient data (age, cholesterol, blood pressure, ECG readings) but lack the in-house data science teams to turn it into predictive insights. The result is reactive care: diagnoses come late, interventions are expensive, and outcomes suffer. The business need is to clearly identify high-risk patients earlier, with a model anyone on the team can build and maintain, not just data scientists.

Description:

In this project, you'll use **Amazon SageMaker Canvas**: AWS's no-code machine learning service to build, train, and deploy a heart disease risk prediction model without writing a single line of code.

You'll import a real healthcare dataset, clean and analyze it with **SageMaker Data Wrangler**, automatically train a model using **SageMaker Autopilot**, evaluate its performance, and deploy it as a

live **SageMaker endpoint** that returns predictions on new patient data in real time. By the end, you'll have a working ML pipeline that proves how business teams can use AWS to solve real-world healthcare problems and a portfolio project you can showcase on LinkedIn and in interviews to demonstrate hands-on AWS AI/ML skills.

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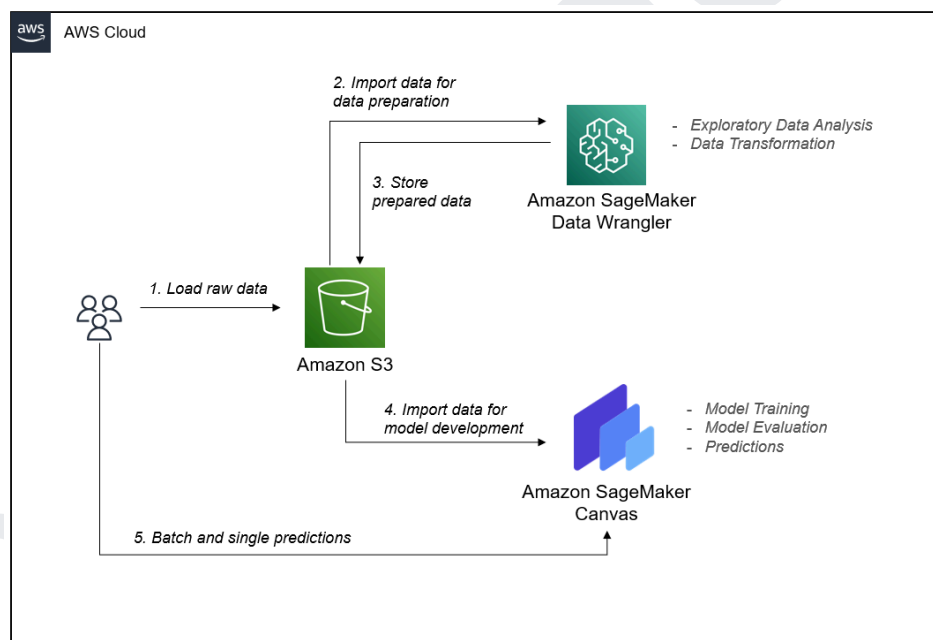
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1 Introduction

Amazon SageMaker Canvas is a **no-code machine learning (ML)** service offered by AWS. It allows business analysts and users with little to no technical expertise in machine learning to easily build, train, and deploy ML models using a **drag-and-drop interface**.

This is an ideal tool for users who want to make predictions based on their data without needing to learn complex programming or data science concepts.

In this project, the goal is to **predict heart disease** using a **public dataset**. By leveraging SageMaker Canvas, users can import the dataset, prepare it, train a model, evaluate its performance, and deploy the model—all without any coding.



This process is guided by a simple, visual interface, making it highly accessible for beginners.

The project involves:

1. **Importing the dataset** into SageMaker Canvas.
2. **Cleaning and preparing** the data for training.

3. **Training a machine learning model** automatically within the Canvas interface.
4. **Evaluating the model** based on key performance metrics.
5. **Deploying the model** for real-time predictions.

This tool essentially empowers business users to incorporate machine learning into their workflows without the steep learning curve traditionally associated with data science and ML.

2 Overview & Key Concepts

In this section, we'll cover key concepts used in this guide, including an overview of what's covered in this guide.

2.1 No-Code Machine Learning

This concept refers to building machine learning models without writing any code. With tools like SageMaker Canvas, users can focus on the data and the results rather than the underlying programming and algorithms.

The tool automates the complex aspects of ML, such as model selection, training, and evaluation.

2.2 Data preparation

Data preparation is one of the most important steps in any machine learning project. In this context, SageMaker Canvas provides easy-to-use tools to clean and preprocess data.

This may include handling missing values, transforming variables, encoding categorical data, and more.

2.3 Model Training

Model training involves feeding the prepared data into a machine learning algorithm to learn from it.

SageMaker Canvas automates this process by selecting the appropriate model based on the dataset and training it to predict outcomes (in this case, whether or not a person is likely to have heart disease).

2.4 Model Training

After training, it's crucial to evaluate the model's performance to determine its effectiveness.

In SageMaker Canvas, the tool automatically evaluates the model using common metrics like accuracy, precision, recall, and F1 score.

This helps users understand how well the model can predict heart disease from new data.

2.5 Model Deployment

Deployment is the process of making the trained model available for use in a real-world setting.

SageMaker Canvas provides a simple way to deploy the model to a cloud endpoint, allowing it to make predictions based on new input data.

3 Documentation

1. Amazon SageMaker

<https://aws.amazon.com/pm/sagemaker/>

2. Amazon SageMaker Canvas

<https://docs.aws.amazon.com/sagemaker/latest/dg/canvas.html>

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4 Prerequisite

1. A Paid AWS Account: If you don't have an AWS Account, then please check Activity **Guide: Create AWS Account** [Here](#).

The screenshot displays the 'AWS for Beginners' course interface. On the left, a sidebar lists the course content, with 'Lab 1: Create an AWS Cloud Account' highlighted in a red box. The main area shows a video player for 'Lab 1: Create an AWS Cloud Account' with a red box around the title. Below the video player, a red box highlights the first resource: '1. Step-by-Step Guide: [Create AWS Account & Access Console](#)'. A second red box highlights the second resource: '2. Related / References' with a link to <https://aws.amazon.com/free/>.

2. To increase quota, please follow [Activity Guide: Requesting AWS Service Quota Increases For EC2](#) Instances

AWS AIF-C01 AI Practitioner The... 0%

Module 0: Start Here

Module 1: Amazon Bedrock and Ge...

Module 2: Prompt Engineering

Module 3: Amazon Q

Module 4: AI, ML, Deep Learning, a...

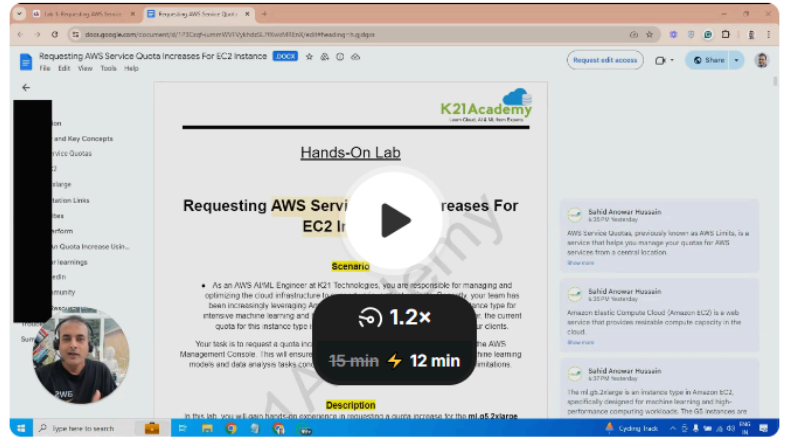
Module 5: AWS Managed AI Services

Module 6: Amazon SageMaker

Presentations

L1 : Intro to Sagemaker

Lab 1: Requesting AWS Service Quota Increases



Refer to Step by Step Guide >>> [Requesting AWS Service Quota Increases For EC2 Instance](#)

3. [Download the Heart Disease Dataset](#) => We will use this dataset for data processing, training a machine learning model, and making predictions.

5 Cost to Perform this Lab

Cost Estimate: There will be charges for using SageMaker Canvas based on the usage beyond the Free Tier limits.

Pricing Breakdown:

- **SageMaker Canvas Free Tier:** 160 hours per month.
- **Beyond Free Tier:** ~\$1.9 per hour for Canvas Workspace instances.

For the current lab session, the cost breakdown based on usage:

- **Usage:** 2.081 hours
- **Cost:** 2.081 hours * \$1.9 per hour = **\$3.95**

SageMaker			
US West (Oregon)			USD 4.52
Amazon SageMaker CreateVolume-Gp2			USD 0.00
Amazon SageMaker CreateVolume-Gp3			USD 0.04
\$0.112 per GB-Mo of provisioned ML storage		0.321 GB-Mo	USD 0.04
Amazon SageMaker Invoke-Endpoint			USD 0.00
Amazon SageMaker RunInstance			USD 4.48
\$1.515 per Hosting ml.g5.2xlarge hour in US West (Oregon)			0.287 Hrs
\$1.9 per Hrs for Canvas:Workspace Instance (Session-Hrs):RunInstance			2.081 Hrs
			USD 3.95

Recommendation:

Be sure to delete all resources after completing the lab to avoid any unexpected charges.

For More Pricing Details:

<https://aws.amazon.com/sagemaker/pricing/>

6 Create a Domain to Launch Amazon SageMaker Canvas

In this section, we'll set up a **SageMaker Domain** using the AWS Console.

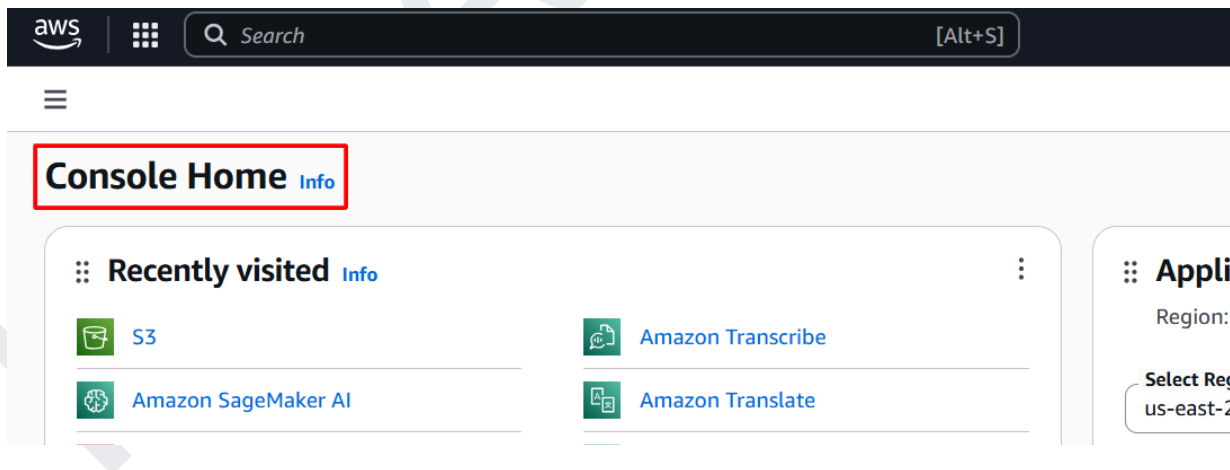
A SageMaker Domain is a secure workspace where you can manage your machine learning projects. It helps you organize your resources, manage users, and easily connect with other AWS services.

Creating a domain sets up everything you need to run notebooks, train models, and collaborate with your team. It ensures that your work is saved and can be accessed whenever you log in.

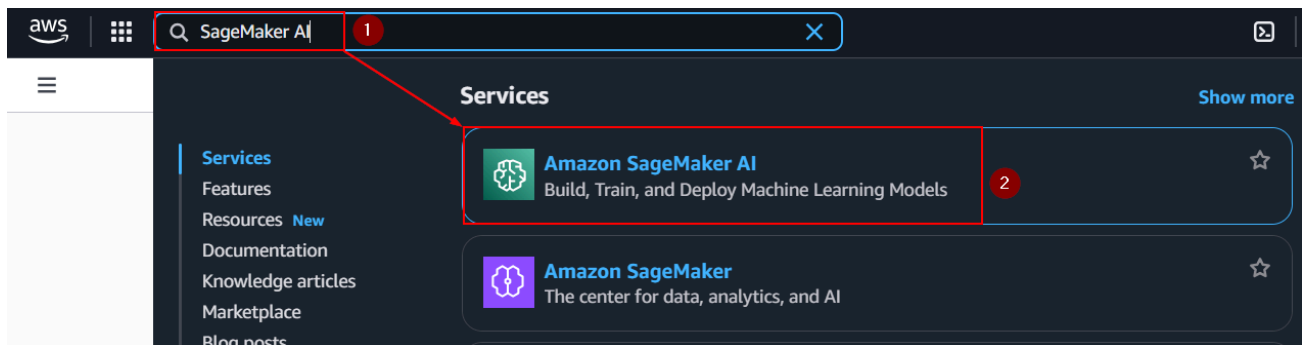
We'll guide you through the steps to create the domain and a user profile, so you can start using SageMaker Canvas without any hassle.

Note: Before starting the lab, please ensure you have reviewed and completed all the requirements listed in the [prerequisites](#) section to avoid any issues during the lab.

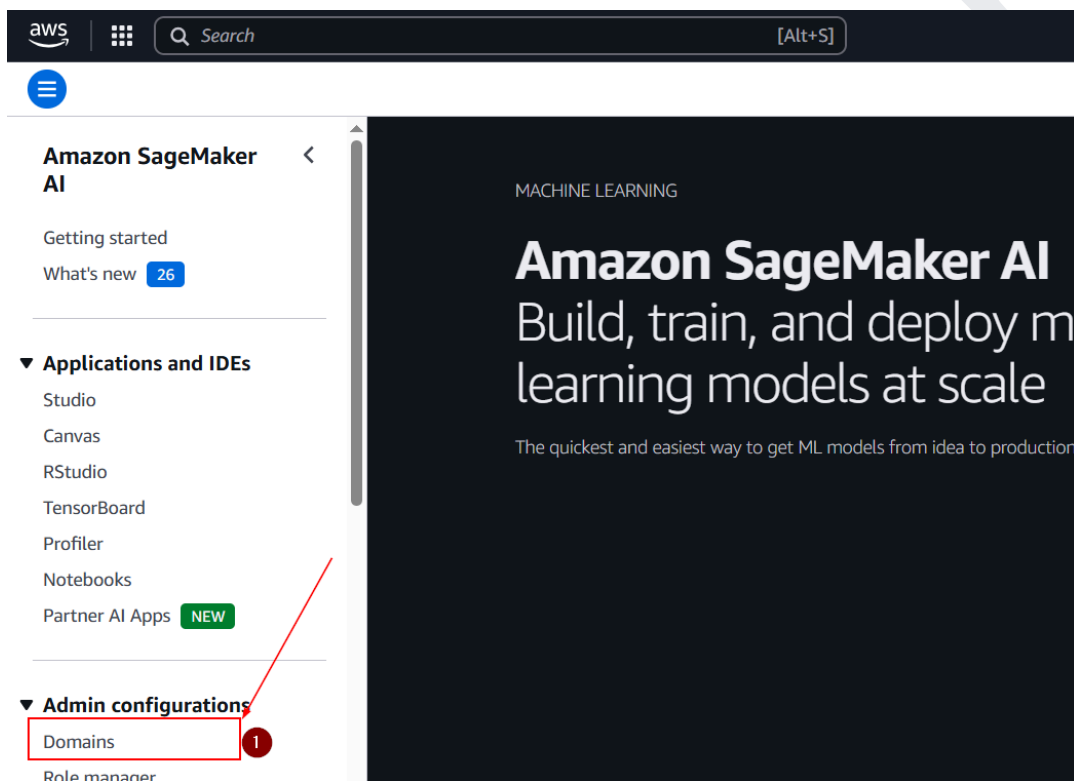
1. Log in to the [AWS Console](#).



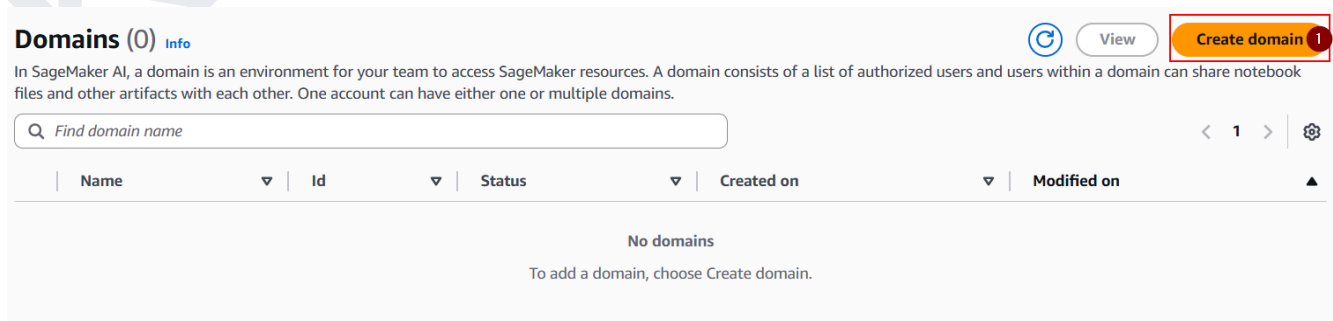
2. Search for **SageMaker AI** and open the service.



3. In the left menu, under **Admin configurations**, click **Domains**.



4. Click **Create domain**.



- Choose **"Set up for single user (Quick setup)."** This will automatically create a domain and user profile for you.

Set up SageMaker Domain

Use SageMaker Domain as the central store to manage the configuration of SageMaker for your organization.

Set up for single user (Quick setup)

Let Amazon SageMaker configure your account, and set up permissions for your SageMaker Domain.

- ✓ New IAM role with AmazonSageMakerFullAccess policy
- ✓ Public internet access, and standard encryption
- ✓ SageMaker Studio - New, and SageMaker Studio Classic integrations
- ✓ Shareable SageMaker Studio Notebooks
- ✓ SageMaker Canvas
- ✓ IAM Authentication

Perfect for single user domains and first time users looking to get started with SageMaker.

Set up for organizations

Control all aspects of account configuration, including permissions, integrations, and encryption.

- ✓ Advanced network security, and data encryption
- ✓ SageMaker Studio - New, SageMaker Studio Classic, RStudio, and Code Editor Based on Code-OSS, Visual Studio Code Open Source integrations
- ✓ SageMaker Studio Projects, and Jumpstart
- ✓ SageMaker Canvas, and Amazon services integrations
- ✓ IAM, or IAM Identity Center (successor to AWS SSO)

Better for admins with large user groups, but you can always update your account configuration settings later if you want to do a quick setup now.

[Other domain setup options](#)

2 **Set up**

- Once the quick setup begins, you will see the message:

"Preparing SageMaker Domain. Please do not close this browser tab. This one-time setup process may take several minutes to complete."

- Please be patient and avoid closing the tab until the setup finishes.

Amazon SageMaker AI > Domains > Domain: QuickSetupDomain-20250514T145797

Preparing SageMaker Domain. Please do not close this browser tab.

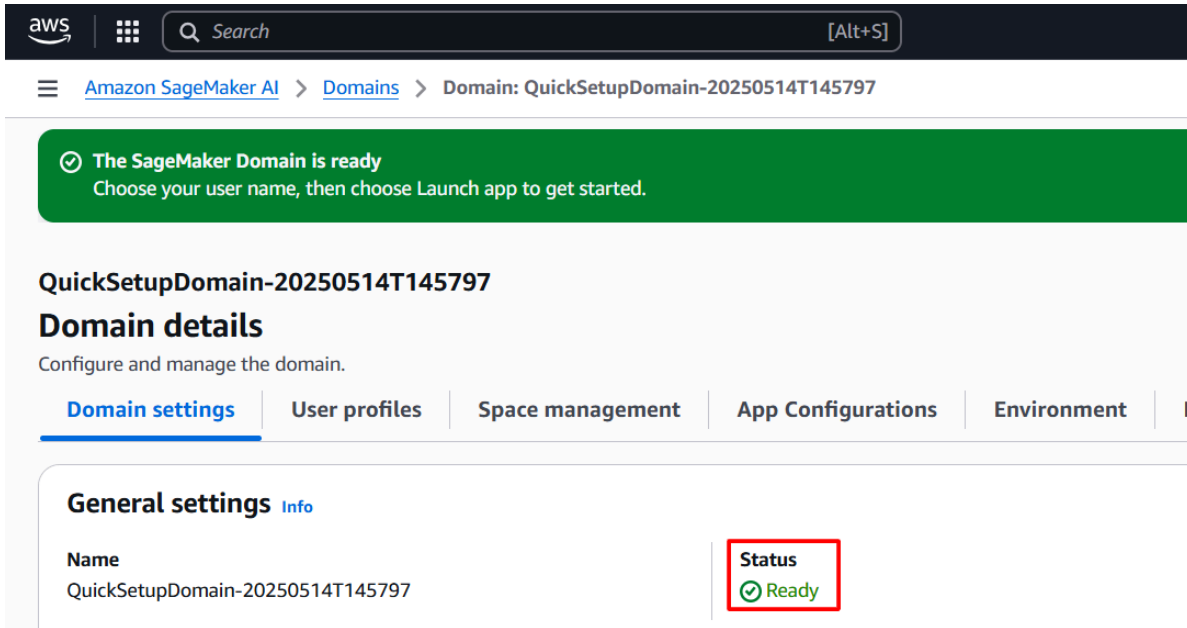
We are configuring the resources needed by the domain. This is a one-time configuration and may take a few minutes.

QuickSetupDomain-20250514T145797

Domain details

Configure and manage the domain.

- The quick setup automatically creates a SageMaker domain and user profile using default settings. As an administrator, you can view and modify these domains and user profiles as needed.



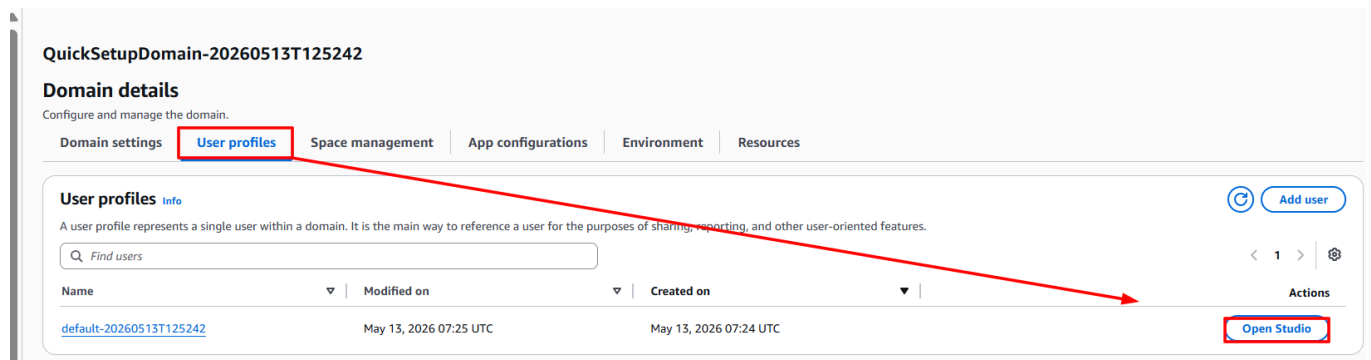
The screenshot shows the AWS SageMaker Domains console. At the top, a green banner states: "The SageMaker Domain is ready. Choose your user name, then choose Launch app to get started." Below this, the domain name "QuickSetupDomain-20250514T145797" is displayed. Under "Domain details", there is a navigation bar with tabs: "Domain settings" (selected), "User profiles", "Space management", "App Configurations", "Environment", and "R". Under the "General settings" section, the "Name" is "QuickSetupDomain-20250514T145797" and the "Status" is "Ready", which is highlighted with a red box.

The Quick Setup is now complete. You can begin using SageMaker Canvas and explore its features.

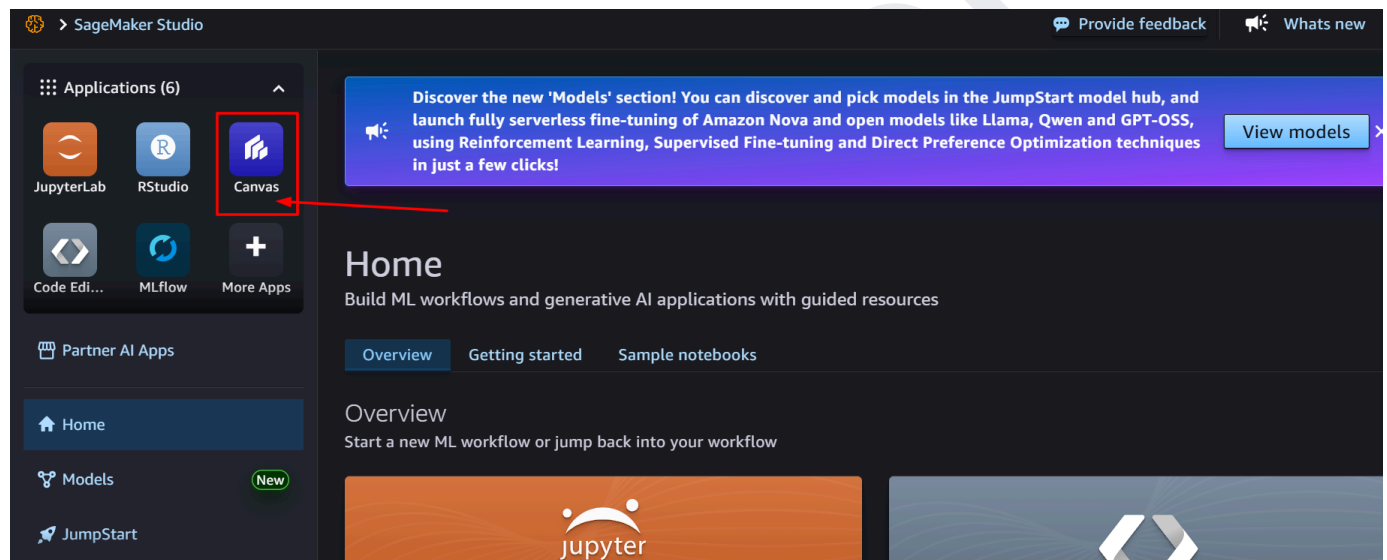
6.1 Launch Amazon SageMaker Canvas

Amazon SageMaker Canvas is a **no-code machine learning (ML) service** that enables business analysts and users with no data science expertise to easily build, train, and deploy machine learning models.

1. On the Amazon SageMaker dashboard, click on **Domains**. Then, select the domain that you recently created.
2. Click on the **user profiles** under **Domain details**, then click **Launch**, and finally select **Canvas** to open SageMaker Canvas.



3. It may take approximately 5 minutes for Canvas to start.



SageMaker Studio > Canvas

Applications (6)

- JupyterLab
- RStudio
- Canvas
- Code Edi...
- MLflow
- More Apps

Partner AI Apps

Home

Models

JumpStart

Canvas Private

Build and use ML and generative AI models - no code required

Run Canvas

Status ?

Stopped

No-code ML and generative AI journey

- Prepare data**
Explore and transform data with natural language or a visual interface.
- Predict outcomes**
Generate business insights with custom, ready-to-use, or generative AI models.
- Train models**
Train custom models, or fine-tune use case.
- Automate workflow**
Deploy models to production

SageMaker Studio > Canvas

Applications (6)

- JupyterLab
- RStudio
- Canvas
- Code Edi...
- MLflow
- More Apps

Partner AI Apps

Home

Canvas Private

Build and use ML and generative AI models - no code required

Stop Canvas

Open Canvas

Status ?

Running

No-code ML and generative AI journey

- Prepare data**
Explore and transform data with natural language or a visual interface.
- Train models**
Train custom models, or fine-tune generative use case.

7 Data Preparation with SageMaker Data Wrangler

Before we start building the model, let's discuss **data preparation**. This step is often the most time-consuming part of any machine learning project.

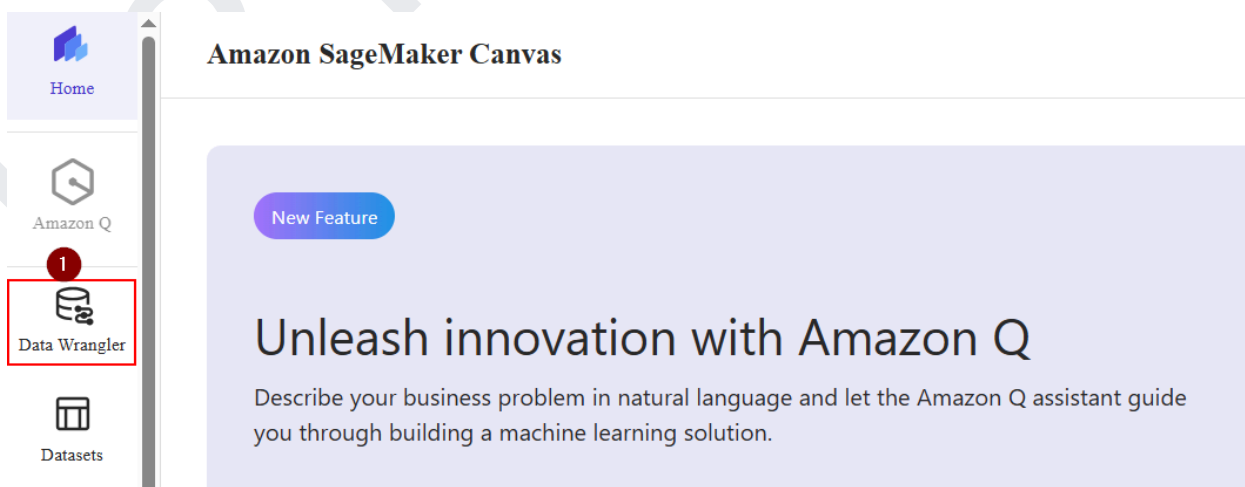
Amazon SageMaker Data Wrangler helps to speed up this process significantly. It's a powerful tool that can transform the time it takes to clean and prepare your data—whether it's tabular, image, or text data—from weeks to just a few minutes.

In our heart disease prediction project, we will leverage **SageMaker Data Wrangler** to **efficiently assess the quality of our dataset and get it ready for training**. This tool will help us quickly identify data issues and ensure that the features are prepared for machine learning, all without writing any code.

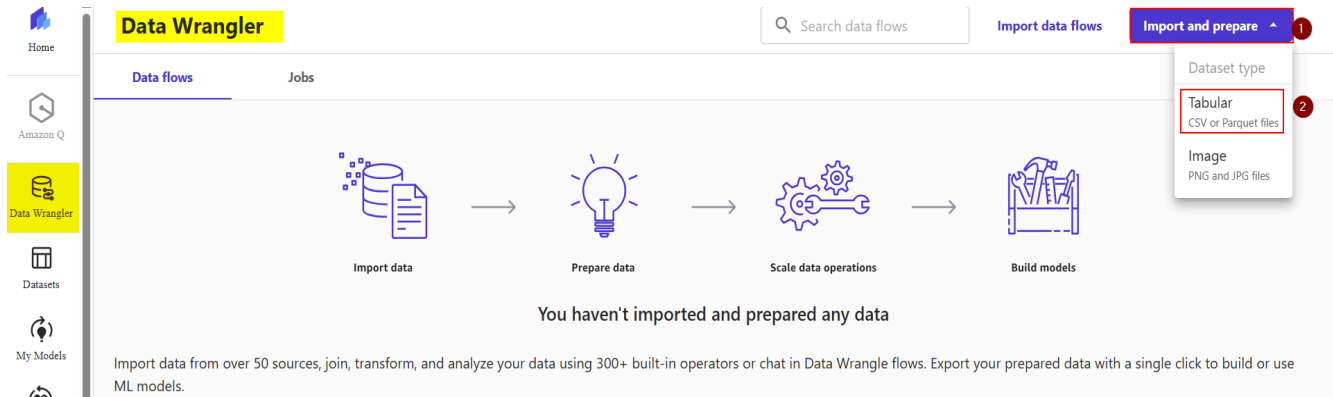
We'll start by uploading and importing our dataset into **SageMaker Data Wrangler**. Once the data is loaded, we'll generate a **Data Quality and Insights Report**.

This report will automatically analyze the data for common issues such as missing values, duplicate entries, and any anomalies like outliers or class imbalances.

1. In the **SageMaker Canvas** interface, navigate to the **left-hand menu** and click on **Data Wrangler** to open the data preparation tools for your project.



- In **SageMaker Data Wrangler**, click on **Import and Prepare** in the top right corner. Then, select **Tabular** as your dataset type to upload heart-disease-dataset.csv for further processing.



- Set the data source to **Local upload**. Then, click on **Select files from your computer**, choose the **heart-diseasedataset.csv** file, and click **Next**.

Import tabular data

Select a data source: Local upload 1

Upload files to import

Drag a CSV or Parquet file here

or

Select files from your computer 2

- Click on **Import**.

Import tabular data

← Previewing 1 file Showing the first 100 rows Import settings

📘 If your data has special character delimiters, use the advanced import settings to specify a custom delimiter. [Learn More](#)

heart-disease-dataset.csv Delete

age	sex	cp	trestbps	chol
63	1	3	145	233
37	1	2	130	250
41	0	1	130	204
56	1	1	120	236
57	0	0	120	354
57	1	0	140	192

Import settings

Settings apply to all imported files. [Learn more](#)

Dataset name *

heart-disease-dataset.csv

Sampling

Sample your dataset for faster exploration. Your full dataset will be used for data export or model build. [Learn more](#)

Sampling method *

Random

Sample size

50000

1 50k 100k 150k 200k (Recommended)

Cancel Back Import

- In the **Data types** section, click on the **plus (+) icon**, then select **Export**. From the options, click on **Export data to Canvas dataset** to transfer the prepared data into SageMaker Canvas for model building.

Data Wrangler: Data flow > New data flow 2025-6-23 5:18:55 AM.flow Add data

Step 2: Data types Data flow Data Analyses

Validation complete 0 errors Done

Source

Local File Upload: heart-disease...

Data types

Transform: heart-disease-dataset.csv

+ Add transform

Get data insights
Identify data quality issues and get recommendations.

Chat for data prep
Use GenAI to explore and prepare your data.

Combine data >

Create model
Export data and start model building in Canvas.

Export
Export your data and create job.

Export data to Canvas dataset
Build ML models to generate insights.

Export data to Amazon S3
To use prepared data outside of Canvas.

Export via Jupyter notebook >
View and run your transforms in Python code.

Add destination
Add a destination node in Amazon S3 to store your data when exporting or creating a job.

Run validation

- In the **Dataset details** window, name the dataset "**Heart Disease Data**". After that, click on **Export**

Export to Canvas dataset ✕

Create a Canvas dataset, and then build your own ML model or get predictions from one of our ready-to-use models.

Dataset details

Dataset name * 1

Heart Disease Data

Use only letters, numbers, spaces, dashes, colons, and underscores up to 64 characters.

☒ Process entire dataset i

Cancel

2
Export

7. The **Heart Disease Data** dataset has been successfully exported

Step 2: Data types

Data flow | Data | Analyses

Validation complete 0 errors [Done](#) [Run validation](#)

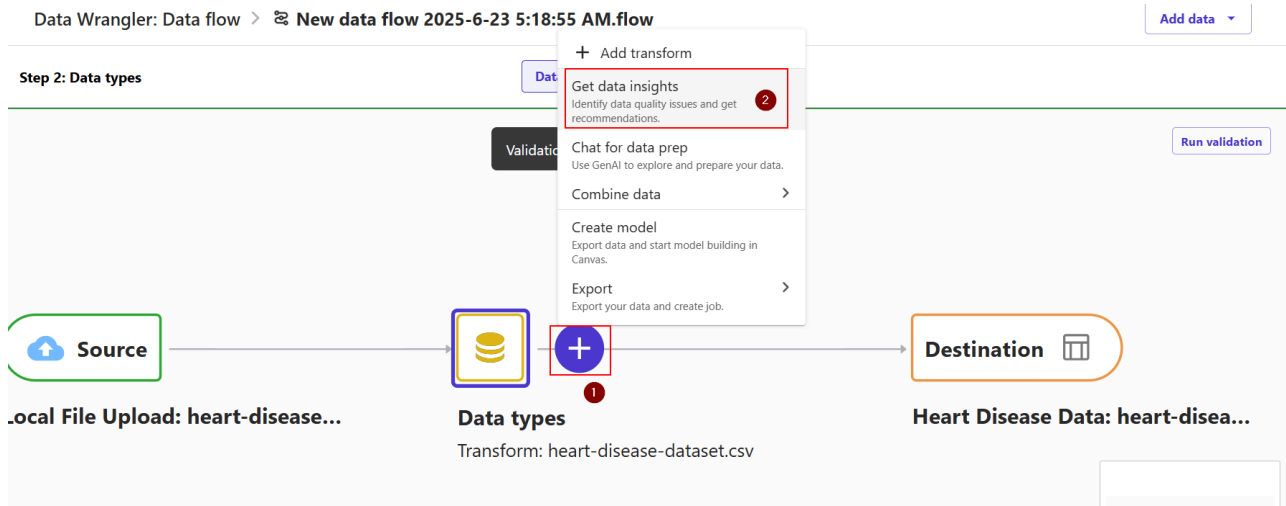
Source → Data types (+) → Destination

Local File Upload: heart-disease... | Data types: Transform: heart-disease-dataset.csv | Heart Disease Data: heart-disea...

Columns: 14 Rows: 303

✓ Heart Disease Data dataset exported successfully. [Create model](#) ✕

8. In the **Data types** section, click on the **plus (+) icon**, then select **Get data insights** to automatically analyze the dataset for data quality issues and receive recommendations.



9. Click on **Create**.

Data Wrangler: Data flow > New data flow 2025-6-23 5:18:55 AM.flow > heart-disease-dataset.csv

Step 2: Data types

Data flow Data Analyses Create model

Data Quality And Insights Report +

Data Quality And Insights Report: 6-23-2025_5:35:58_AM

No Preview available

Create analysis

Data Quality And Insights Report

Analysis name: 6-23-2025_5:35:58_AM

Target column: Select...

Data size * ⓘ

☒ Sampled dataset

☐ Full dataset

Clear Create

10. The **Data Quality and Insights Report** provides a detailed analysis of the heart disease dataset, summarizing key statistics and identifying potential issues.

Summary

Dataset statistics

Key	Value			Feature type	Count
Number of features	14			numeric	10
Number of rows	303			categorical	0
Missing	0%			text	0
Valid	100%			datetime	0
Duplicate rows	0.66%			binary	4
				unknown	0

High Priority Warnings

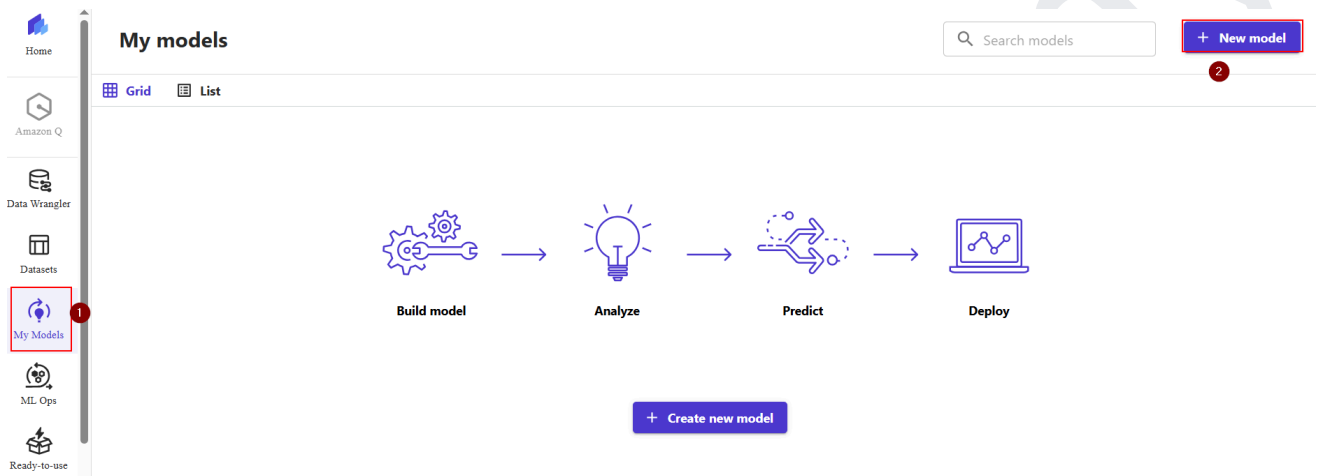
No high severity warnings were detected in the data.

This analysis helps us make better decisions for data cleaning, feature engineering, and choosing the right model in SageMaker Canvas. We are now ready to build a predictive model for heart disease.

8 Build a Machine Learning Model using Amazon SageMaker Canvas

In this section, we will go through the steps to **build a machine learning model using Amazon SageMaker Canvas to predict the risk of heart disease.**

1. In **Amazon SageMaker Canvas**, on the **left-hand menu**, click on **My Models**. Then, click on **New Model** to start creating a new machine learning model.



2. Name the model **"Heart Disease Risk Prediction"**, select **Predictive analysis** as the problem type, and then click **Create**.

Create new model

Model name

Model name

Heart Disease Risk Prediction

Use only letters, numbers, and underscores, up to 32 characters.

Problem type

Select the problem type you want the model to solve.

☒ Predictive analysis

Build models using tabular datasets to predict single or multiple categories as well as regression and time-series forecast problems.

☐ Image analysis

Build models using image datasets to predict single or multiple categories for image classification problems.

☐ Text analysis

Build models using tabular datasets to predict single or multiple categories for text classification problems.

☐ Fine-tune foundation model

Customize a foundation model on your data to improve its performance for a specific task or domain.

Cancel

Create

3. Click on **Heart Disease Data** from the list, then click on **Select dataset** to proceed.

My models > Heart Disease Risk Prediction > Version 1 + Create new version ↺ ⋮

Select Build Analyze Predict Deploy

Select dataset + Create dataset

You can import a tabular dataset or choose one that has already been imported. Your dataset must contain at least one input column and a target column.

🔍 Search datasets in Canvas

All Joined

Name	Columns	Rows	Cells	Created	Status
☒ Heart Disease Data 1 V1	14	303	4,242	06/23/2025 5:30 AM	Ready
☐ canvas-sample-loans-part-1.csv V1	19	1,000	19,000	06/23/2025 5:06 AM	Ready
☐ canvas-sample-loans-part-2.csv V1	5	1,000	5,000	06/23/2025 5:06 AM	Ready
☐ canvas-sample-retail-electronics-forecasting.csv V1	6	40,500	243,000	06/23/2025 5:06 AM	Ready
☐ canvas-sample-databricks-dolly-15k.csv V1	2	10,879	21,758	06/23/2025 5:06 AM	Ready

2 **Select dataset**

4. Amazon **SageMaker Canvas** automatically selects the appropriate model type based on the target column you choose. Once you select the **Target column**, it will recommend a model type. If needed, you can adjust the model type to suit your specific requirements.
5. Select the target column as **CP (chest pain)**, then click on **Quick Build** to automatically start the model-building process.

My models > Heart Disease Risk Prediction > Version 1

+ Create new version

Select

Build

Analyze

Predict

Deploy

Select a column to predict

Choose the target column. The model that you build predicts values for the column that you select.

Target column

cp

Value distribution



Model type

SageMaker Canvas automatically recommends the appropriate model type for your analysis.

3+ category prediction

Your model classifies cp into 3 or more categories.

Configure model

Quick build

Preview model

Heart Disease Data

Full dataset: 303 rows

Manage columns

Manage rows

Time series

View all

Data visualizer

Column name	Data type	Feature type	Missing	Mismatched	Unique	Mode
trestbps	123 Numeric	-	0.00% (0)	0.00% (0)	49	120
thalach	123 Numeric	-	0.00% (0)	0.00% (0)	91	162

Note: The **Quick Build** prioritizes speed over accuracy and takes about 15 minutes to complete. If you need higher accuracy, you can choose the **Standard Build**, which takes 2-4 hours. For this project, we're opting for the **Quick Build** to get faster results.

6. Your model is being created. Quick build usually takes 2–15 minutes.

My models > Heart Disease Risk Prediction > Version 1

+ Create new version

Select

Build

Analyze

Predict

Deploy

Model overview

Your model is being created. Quick build usually takes 2–15 minutes. You can now leave this view.

Expected build time

2–15 minutes

Build type

Quick build

Detailed progress

Generating column impact



7. We have successfully built the model. Now, let's analyze the model's outcomes.

My models > Heart Disease Risk Prediction > Version 1

+ Create new version



Select Build **Analyze** Predict Deploy

Model status [Quick build](#)

Accuracy [Optimization metric](#)

56.452%

The model predicts the correct Cp 56.452% of the time.

Predict

Standard build

Deploy

Overview

Scoring

Advanced metrics

Model leaderboard

Column impact [Quick build](#)

Search columns...

1	target	19.528%
2	exang	13.336%
3	chol	11.679%

Impact on prediction

Impact of target on prediction of cp

0

2.07...

0.01...



- The **accuracy** of the model is **56.45%**, meaning the model correctly predicts the target column (**CP - chest pain**) **56.45% of the time**.
- Target Column (CP)** has the **highest impact** on the prediction (19.53%). This means that the chest pain column plays a significant role in determining whether a person is at risk of heart disease.

8. Now that the model is trained, let's use it to predict the risk of heart disease for the dataset.

9. Click on **Predict**.

My models > Heart Disease Risk Prediction > Version 1

+ Create new version

Select Build **Analyze** Predict Deploy

Model status [Quick build](#)

Accuracy [Optimization metric](#)

56.452%

The model predicts the correct Cp 56.452% of the time.

Predict

Standard build

Overview

Scoring

Advanced metrics

Model leaderboard

Column impact [Quick build](#)

Impact of target on prediction of cp

10. To predict the heart disease risk, select **Batch prediction** to generate predictions for the entire dataset. Then, click on **Manual** and select the **Heart Disease Data** dataset.

My models > Heart Disease Risk Prediction > Version 1

Select

Build

Analyze

Predict

Deploy

Predict target values

Batch prediction 1

Single prediction

Generate predictions for an entire dataset. ⓘ

Manual 2

Automatic

Predictions

All Jobs

Configuration

Filter by configuration name: All ▼

☐	Job name	Created	Input dataset	Prediction type

11. After a successful prediction, the status will change to **Ready**.

Select dataset for predictions

To make predictions on a dataset, select it or import it. The dataset that you select must have the same number of feature columns as the training dataset. ⓘ [+ Create dataset](#)

Search datasets in Canvas

Name		Columns	Rows	Cells	Created	Status
☒ Heart Disease Data	V1	14	303	4,242	05/13/2026 1:12 PM	Ready
☐ canvas-sample-maintenance.csv	V1	9	1,000	9,000	05/13/2026 1:00 PM	Incompatible ⓘ
☐ canvas-sample-loans-part-1.csv	V1	19	1,000	19,000	05/13/2026 1:00 PM	Incompatible ⓘ
☐ canvas-sample-shipping-logs.csv	V1	12	1,000	12,000	05/13/2026 1:00 PM	Incompatible ⓘ
☐ canvas-sample-housing.csv	V1	10	1,000	10,000	05/13/2026 1:00 PM	Incompatible ⓘ
☐ canvas-sample-retail-electronics-forecasting.csv	V1	6	40,500	243,000	05/13/2026 1:00 PM	Incompatible ⓘ
☐ canvas-sample-loans-part-2.csv	V1	5	1,000	5,000	05/13/2026 1:00 PM	Incompatible ⓘ

Close

Generate predictions

12. Click the **three dots** and select **Preview** to view the probability of getting heart disease

Predictions

[All Jobs](#) Configuration

Filter by configuration name:

 Send to Amazon QuickSight

☐	Job name	Created	Input dataset	Prediction type	Configuration name	Rows	QuickSight	Status
☐	batchInfer-Heart Disease Risk f	06/23/2025 6:32 AM	Heart Disease Data V1	Manual		303	Not Sent	Ready  1

batchInfer-Heart Disease Risk Prediction-Heart Disease Data-1750640565 predictions ready View

batchInfer-Heart Disease Risk Prediction-Heart Disease Data-1750641166

Prediction (cp)	Probability	age	sex	trestbps	chol	fbs	restecg
3	91.7%	63	1	145	233	1	0
2	96.2%	37	1	130	250	0	1
1	96.8%	41	0	130	204	0	0
1	96.5%	56	1	120	236	0	1
0	90.2%	57	0	120	354	0	1
2	82.3%	57	1	140	192	0	1
1	88.0%	56	0	140	294	0	0
1	85.6%	44	1	120	263	0	1
2	95.7%	52	1	172	199	1	1
2	96.7%	57	1	150	160	0	1

 Send to Amazon QuickSight  Download

- Each **Prediction (cp)** represents a class or category, with the following possible values:
 - 0**: No risk or healthy
 - 1**: Low risk
 - 2**: Medium risk

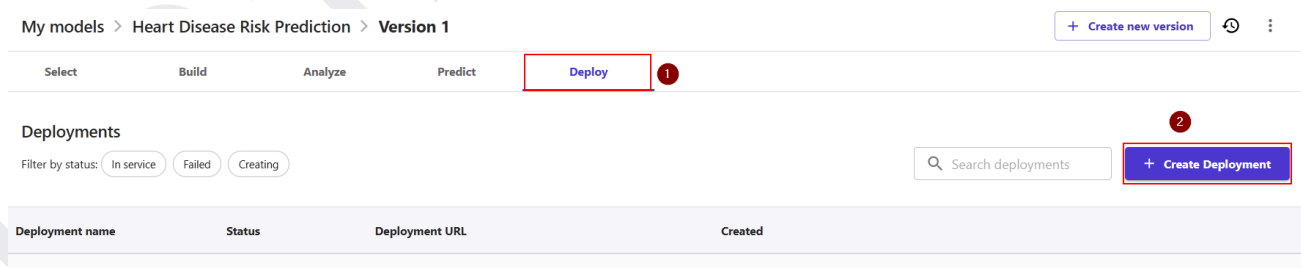
- **3: High risk**
- The **Probability** is the model's confidence (expressed as a percentage) that the given data point belongs to the predicted class (indicated in the **Prediction (cp)** column). For example:
 - A prediction of **3** (High risk) with a **Probability** of **91.7%** means the model is 91.7% confident that the person has a high risk of heart disease.
 - A prediction of **2** (Medium risk) with a **Probability** of **96.2%** means the model is 96.2% confident that the person has a medium risk.

This probability helps assess how confident the model is about its prediction for each individual in the dataset. Higher probabilities indicate more confidence in the classification.

Congratulations! The heart disease risk prediction has been completed.

8.1 Deploy & Test Machine Learning Model

1. You can easily deploy your model to a SageMaker endpoint from the **Deploy** tab. This allows you to **make predictions outside of the Canvas application** and also test and monitor your model to proactively detect issues like model drift.



Note: If you encounter the error “**You don't have permission to deploy models. Please contact your administrator for support.**” Please refer to the [troubleshooting section](#) for detailed steps to resolve the issue.

You don't have permission to deploy models. Please contact your administrator for support.

 Search deployments

 Create Deployment

2. Click on **Deploy**.

Name: **HeartDiseaseRiskdeployment**

Instance Type: **ml.t2.medium**

Instance count: **2**

and monitor your model to proactively detect issues such as model drift.

Selected model version

Heart Disease Risk Prediction

V1  Ready Created: 05-13-2026-1:22 PM

Deployment type

Real-time 

Deployment name

Deployment name
HeartDiseaseRiskdeployment

Instance type

[Learn about pricing](#) 

ml.t2.medium

Instance count

2

Cancel

Deploy

3. It takes **3-5 minutes** to create the deployment, and the status will change to **In Service** once it's ready.

My models > Heart Disease Risk Prediction > Version 1

Select	Build	Analyze	Predict	Deploy
Deployments				
Filter by status: In service Failed Creating				
Deployment name	Status	Deployment URL	Created	
canvas-new-deployment-06-23-2025...	Creating	https://runtime.sagemaker.us-west-2.amazonaws.com/end...	06/23/25 07:05 AM	

4. Now that the deployment is ready, click on the **three dots** icon and select **Test Deployment** to verify the deployment.

Select

Build

Analyze

Predict

Deploy

Deployments

Filter by status:

In service

Failed

Creating

Search deployment

Deployment name	Status	Deployment URL	Created
canvas-new-deployment-06-23-2025...	In service	https://runtime.sagemaker.us-west-2.amazonaws.com/end...	06/23/25 07:05 AM

1

View details

Test deployment

Update configuration

Delete deployment

5. On the left side, you can see the **age** and **chol** values. Enter some random values and update the prediction.

Age: 68

Chol: 250

Operations: Deployment / canvas-new-deployment-06-23-2025-7-02-AM

[Update configuration](#)

Details **Test deployment**

Modify values to predict **cp** in real time.

Column	Value
age	68
sex	1
trestbps	120
chol	250
fbs	0

cp Prediction

You made 2 changes to the values. Update for a new prediction and feature importance metrics.

0

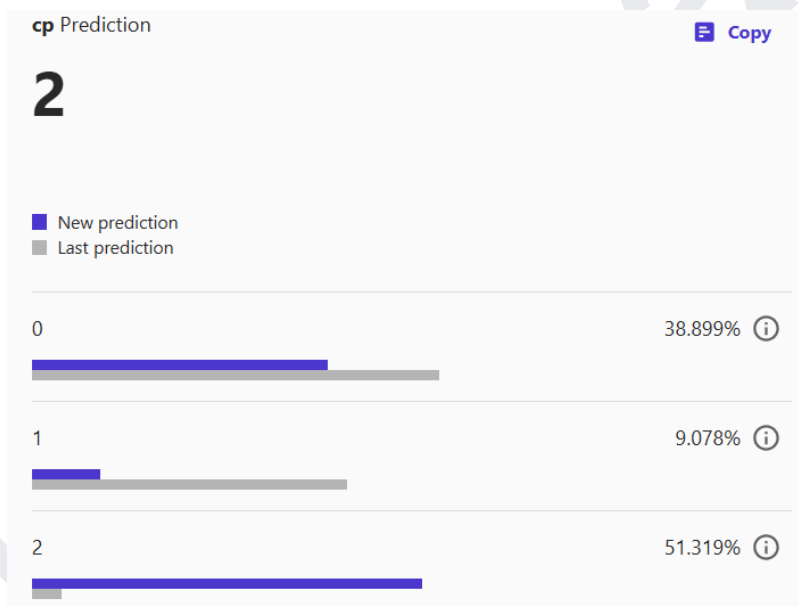
[Update prediction](#)

■ Last prediction

Invocation result

Status	Execution length (ms)	Request time
✓ Successful	205.668	2025-06-23 07:11:41 AM

6. The prediction has been updated to show that, because we increased the values, the chance of **chest pain(cp)** is now higher than before.



We have successfully deployed the machine learning model, allowing it to make predictions on new data without retraining.

9 Share your Learnings on LinkedIn & Community

In this section, you are going to share whatever you learned during this lab on LinkedIn and the Community

Please watch the below 📺 video below to understand why it's important to post & how to post:

<https://www.skool.com/k21academy/classroom/0c24b6af?md=f7975c8e525b4d5d8191e131bdef5740>

LinkedIn for Visibility & Authority ...

40%

LinkedIn: Update Profile, Post Content: Wh...

FAQ: If I Publish, What Current Employer T...

Why it's Imp. to Prepare & Share Not... ✓

Example Post: What & How to Post ... ✓

Useful ChatGPT Prompts ✓

How to create LinkedIn Banner from Canv...

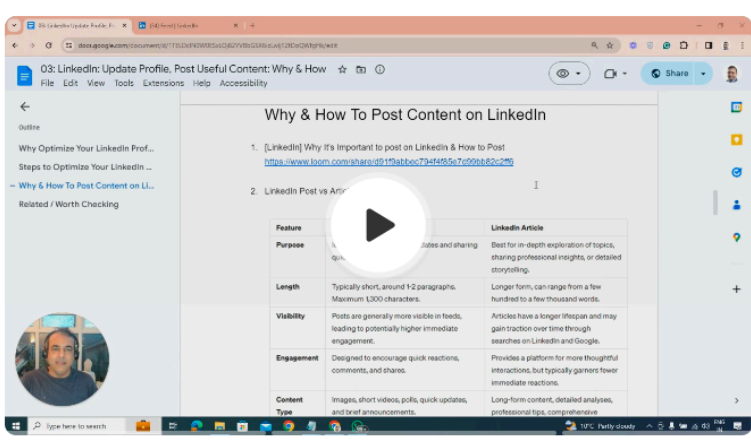
LinkedIn Profile Review ^

Neha: LinkedIn Feedback to Updat... ✓

Deb: Australia Azure + Oracle Cloud ✓

Gagan: LinkedIn Profile Feedback ✓

LinkedIn: Update Profile, Post Content: Why & How ✓ ✎



Why & How To Post Content on LinkedIn

- [LinkedIn] Why It's Important to post on LinkedIn & How to Post
<https://www.linkedin.com/sharing/article?linkid=79448567&url=0c24b6af>
- LinkedIn Post vs Article

Feature	LinkedIn Post	LinkedIn Article
Purpose	Quick updates and sharing	Best for in-depth exploration of topics, sharing professional insights, or detailed storytelling.
Length	Typically short, around 1-2 paragraphs. Maximum 1300 characters.	Longer form, can range from a few hundred to a few thousand words.
Visibility	Posts are generally more visible in feeds, leading to potentially higher immediate engagement.	Articles have a longer lifespan and may gain traction over time through searches on LinkedIn and Google.
Engagement	Designed to encourage quick reactions, comments, and shares.	Provides a platform for more thoughtful interactions, but typically garners fewer immediate reactions.
Content Type	Images, short videos, polls, quick updates, and text announcements.	Long form content, detailed analyses, professional tips, and case studies.

Resources

[Related Document](#)

🧑💻 As you know, building a strong professional profile is essential in today's job market. One way to showcase your skills and knowledge is by sharing your labs and projects on LinkedIn.

Your LinkedIn profile is a powerful tool in your job search. Sharing your labs and projects is just one way to demonstrate your expertise and stand out to potential employers. If you don't have LinkedIn, we strongly recommend you create one for yourself.

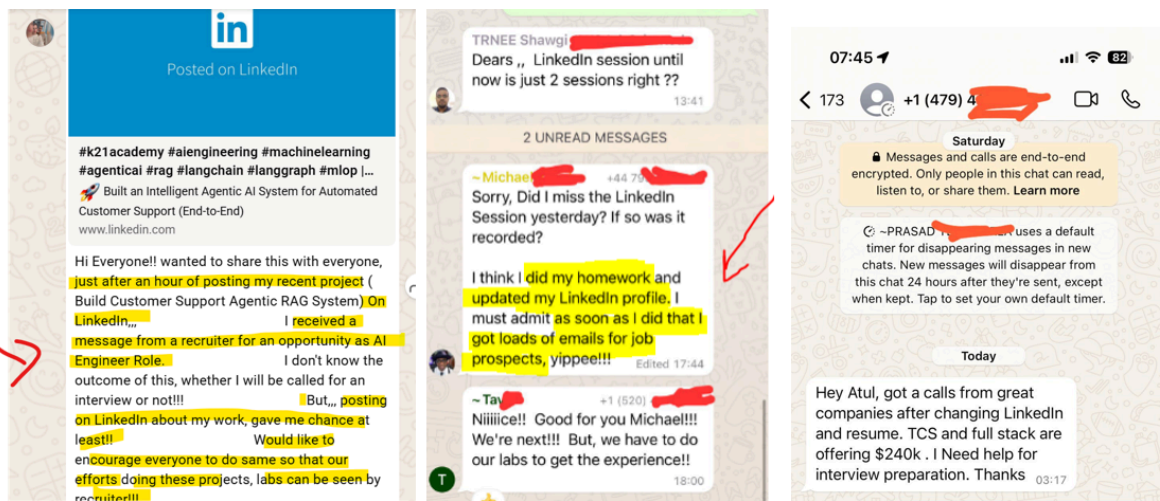
📝💻 By doing so, you can demonstrate to potential employers or connections that you have hands-on experience in your field and are actively engaged in learning and growing your skills.

9.1 Share Hands-On Lab on LinkedIn to Build Authority & Job Opportunities

This will attract recruiters and potential employers to your profile and increase your Authority & LinkedIn reach.

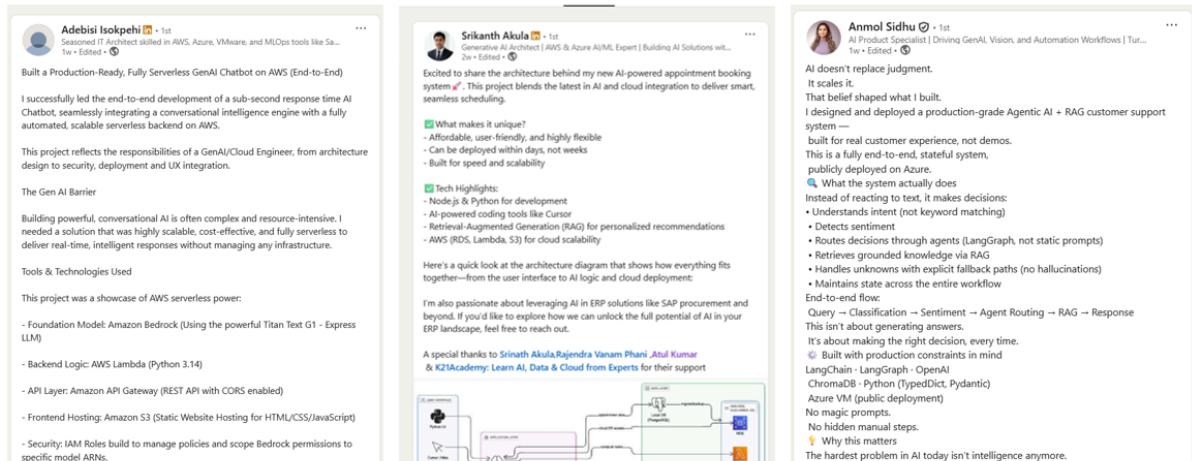
- **Do remember to tag so we can share too, and we'll ask other members to share too (this will increase your post reach)**
 - K21Academy (<https://www.linkedin.com/company/k21academy>) &
 - Atul Kumar (<https://www.linkedin.com/in/atulk21academy/>), as we will further circulate in our network, too, to increase your reach.
- Here is what you can expect
 - Jessy got a message from a recruiter for an AI Engineer Role
 - Michael got loads of emails for Job prospects
 - Prasad got calls from great companies that are offering \$240K Salary

Results You can Expect



- Example Links on How to post
 - Example Posts

Post on LinkedIn for Interview Calls



Example Video & Post

<https://www.skool.com/k21academy/classroom/0c24b6af>

LinkedIn for Visibility & Authority

40%

LinkedIn: Update Profile, Post Content: Wh...

FAQ: If I Publish, What Current Employer T...

Why it's Imp. to Prepare & Share Not...

Example Post: What & How to Post ...

Useful ChatGPT Prompts

How to create LinkedIn Banner from Canv...

LinkedIn Profile Review

Neha: LinkedIn Feedback to Updat...

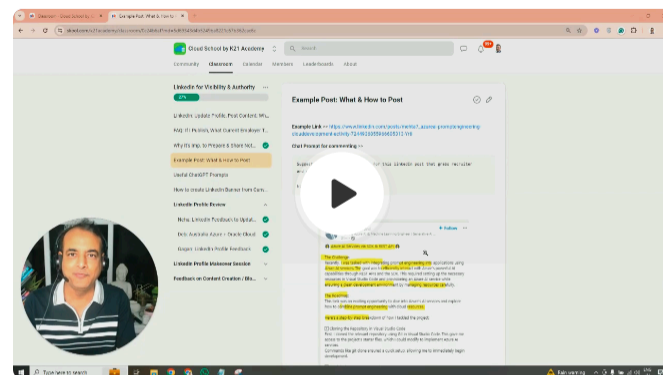
Deb: Australia Azure + Oracle Cloud

Gagan: LinkedIn Profile Feedback

LinkedIn Profile Makeover Session

Feedback on Content Creation / Blo...

Example Post: What & How to Post & Commenting



Example Link >> https://www.linkedin.com/posts/mehta7_azureai-promptengineering-clouddevelopment-activity-724493685596605313-Yrti

Chat Prompt for commenting >>

Suggest useful, engaging comment for this linkedIn post that grabs recruiter and potenial employers attention

<https://www.linkedin.com/>*****

- Check few more linkedIn post examples (do's and don'ts):

<https://www.skool.com/k21academy/classroom/0c24b6af?md=edfe6092fbb84f6bba03ceb439d1e00a>

9.2 Share Your Win on Community via Progress Diary

Share screenshots in Cloud School Community under your **'Progress Diary'**

If you don't know How to Access Your Progress Diary on Skool Portal check this:

<http://k21academy.com/FindYourDiary>

How to Access Your Progress Diary on Skool Portal



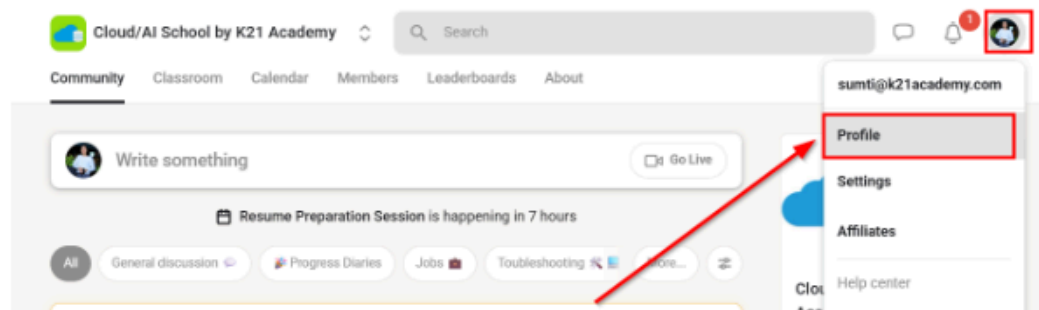
How to Access and Manage Your Progress Diary on Skool Portal

1. Log In to Skool Portal:

- Go to the **Skool** website (or the link provided to you <https://www.skool.com/k21academy>).
- Log in with your credentials (email/username and password).

2. Navigate to Your Profile:

- Once logged in, click on your **profile picture** or **name** at the top right corner of the screen.



This will boost your confidence in the progressive path you are following, and encourage/inspire others to perform these hands-on labs.

Start Here And Helpful Resources ...

0%

Start Here

How To Cancel Your Cloud School Membe...

Create a Progress Diary Post

FAQs

- How to Check Schedule & Live Session ...
- 1. Is there a Mobile App for K21 Acade...
- 2. Progress Tracking for Courses
- 3a. Adjust Video Speed 1.5x or 2.x
- 3b. Screen Quality Adjustment
- 4. How to Create Ubuntu Machine on A...
- 5. How long will we have access to Sko...

Create a Progress Diary Post

This post is designed to help you set up your Progress Diary, which will serve as a log for tracking your learning journey and achievements within the **K21 Academy platform**.



Siram Karthik • just now

Course Name: AWS AI/ML Job Oriented

✅ Lab Name: Heart Disease Prediction Using Amazon SageMaker Canvas: A No-Code Approach

Learning Summary: In this lab, I used Amazon SageMaker Canvas to build a machine learning model for predicting heart disease risk without coding. I first used SageMaker Data Wrangler to generate data insights and quality reports, which helped identify issues like missing values and duplicates. After uploading the dataset, I selected the target column and used the Quick Build option to train the model. The model made predictions based on features like age and cholesterol levels. This no-code approach streamlined the entire process, from data preparation to model deployment.

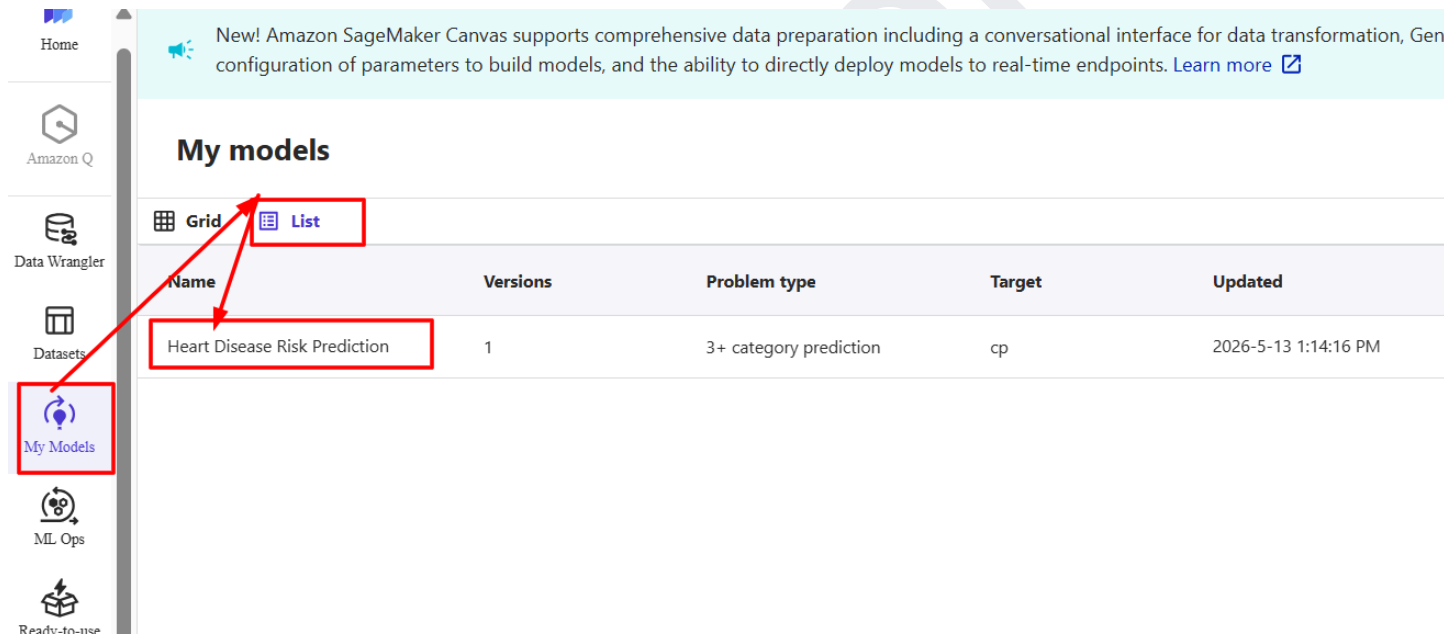
👍 0 Reply

10 Clean up resources

As we have completed the project, it's best practice to delete any resources you are no longer using to prevent incurring unintended charges.

10.1 Delete the deployment and ML Model in Amazon SageMaker Canvas

1. On the Amazon SageMaker Canvas page, click on **My Models** in the left sidebar.
2. Click on the **Heart Disease Risk Prediction** model.



3. Click on **Quick** under the **Build Type** section.

Home

Data Wrangler

Datasets

My Models

ML Ops

Ready-to-use

Versions

Select a version to view details

Version	Status	Build type	Created	Dataset	RMSE	Shared	Model Registry
V1	✓ Ready	Quick 1	10/18/2024 4:11 PM	University...	0.032	--	Not Registered ⓘ ⋮

4. Switch to the **Deploy** section, click on the **three dots**, and select **Delete Deployment**.

Select Build Analyze Predict **Deploy** 1

Deployments

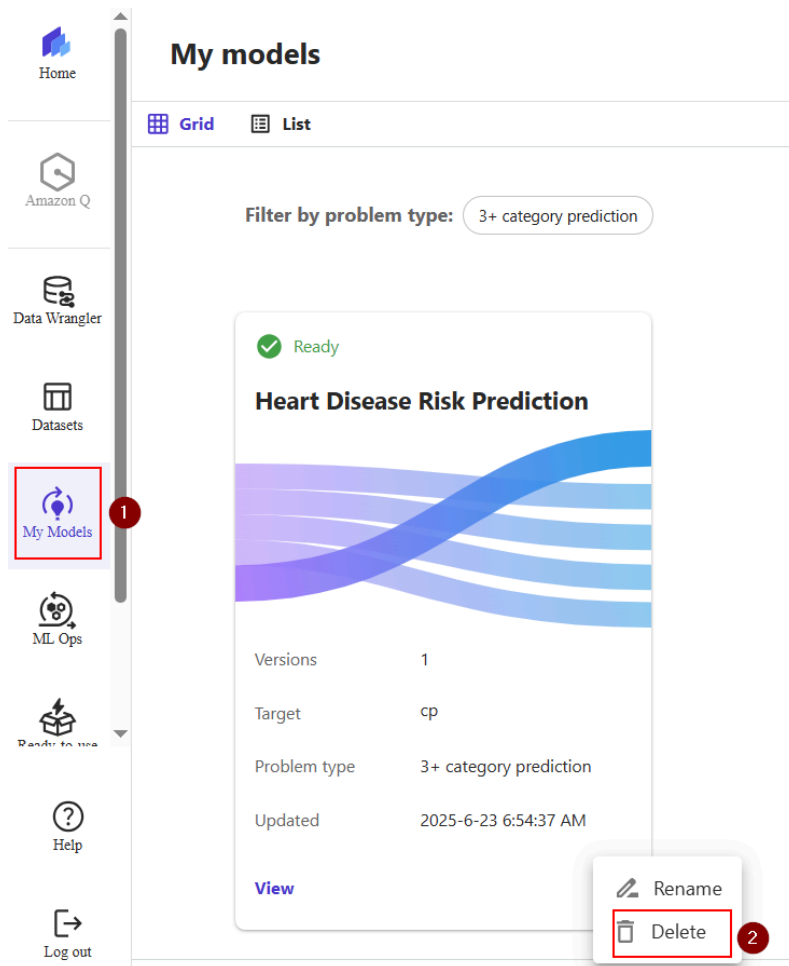
Filter by status: In service Failed Creating Search deployments

Deployment name	Status	Deployment URL	Created
canvas-new-deployment-10-18-2024...	✓ In service	https://runtime.sagemaker.us-west-2.amazonaws.com/end...	10/18/24 04:36 PM

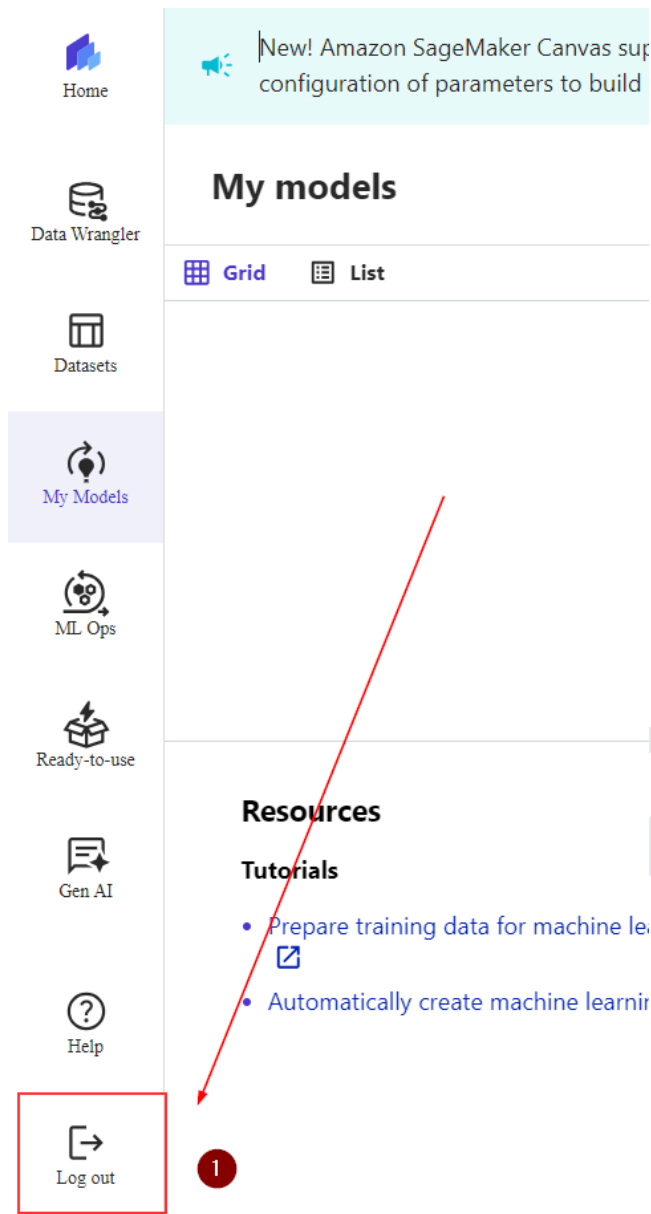
- View details
- Test deployment
- Update configuration
- 2 Delete deployment

5. Then, on the left side, click on **My Models** in the SageMaker Canvas page.

6. Click the **three-dot icon** and delete the **Heart Disease Risk Prediction model**.



7. Log out of Canvas to stop the Canvas service.



10.2 Delete the Models and Endpoints & Configuration

1. The model created by SageMaker Canvas is stored in **Amazon SageMaker >Deployable models> Models**. Deleting the model will remove references to the model artifacts

▼ **Model training & customization**

Training & tuning jobs

JumpStart model hub

Nova Forge **New**

HyperPod clusters

▼ **Training plans**

Training plans

▼ **Deployments & inference**

Deployable models

Endpoints

Endpoint configurations

Batch transform jobs

Shadow tests

Compilation jobs

Inference Recommender

HyperPod inference

► **Model governance**

Models

Search models

Create endpoint Create endpoint configuration Actions Create model

Add/Edit tags 1

Delete

Name	ARN	Creation time
 canvas-model-2025-06-23-01-35-02-386103	arn:aws:sagemaker:us-west-2:442042521384:model/canvas-model-2025-06-23-01-35-02-386103	6/23/2025, 7:05:03 AM

Delete canvas-model-2025-06-23-01-35-02-386103?



 The model you are trying to delete has been created via Studio. Deleting it could cause broken user experience in Studio.

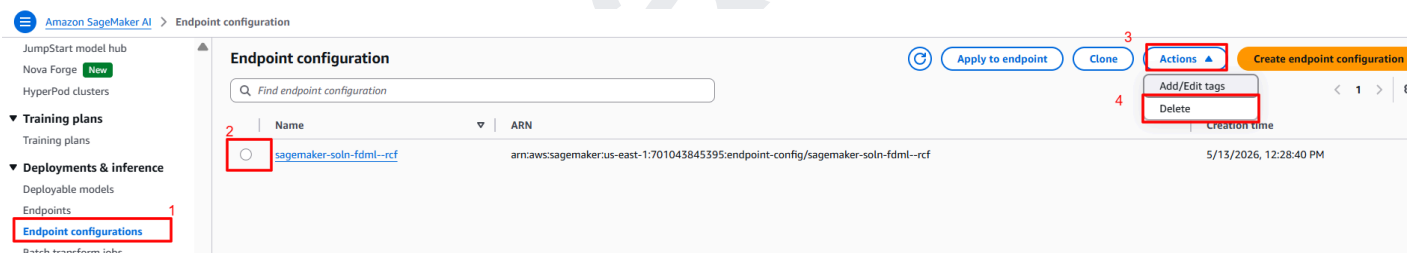
☒ This will permanently delete your model and cannot be undone. This may affect other resources.

Delete

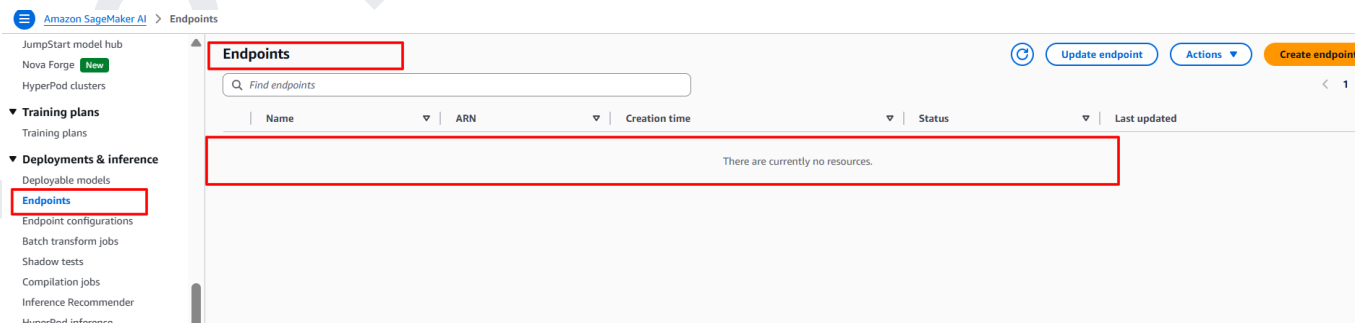
Cancel

2. SageMaker creates an endpoint configuration. You can safely delete this as well:

- Navigate to **Amazon SageMaker > deployment & Inference > Endpoint Configurations**.
- Delete the configurations associated with your project.

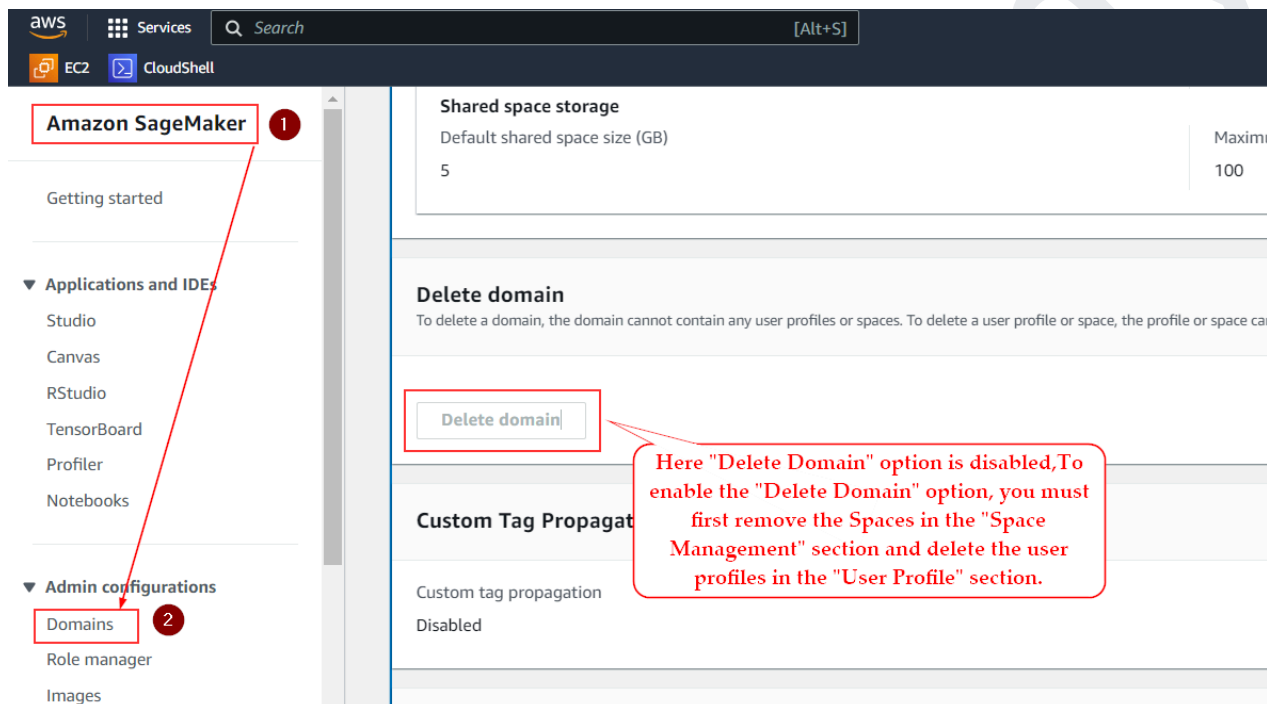


Important: Delete Endpoints from here if any

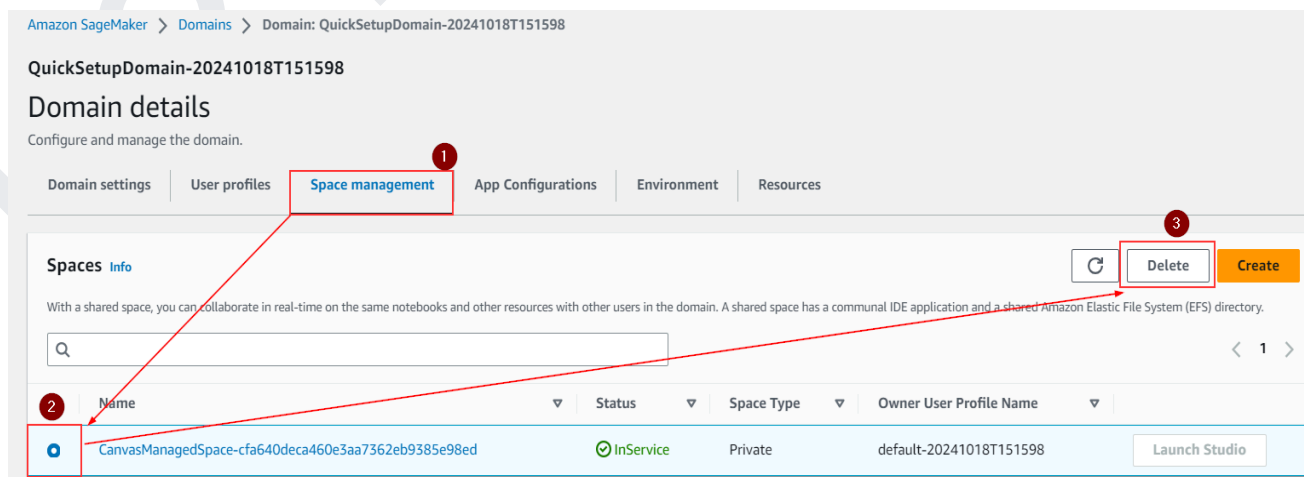


10.3 Delete Amazon SageMaker Domain

1. Delete the domain by navigating to the AWS Management Console, searching for **Amazon SageMaker**, and clicking on **Domains**
 - Here, the **delete domain option is disabled**. To enable the **"Delete Domain"** option, you must first remove the **Spaces in the "Space Management"** section and **delete the user profiles in the "User Profile"** section.



- In the **Domain section**, navigate to **Space Management** and **delete all the spaces**.



Delete space

Delete space
CanvasManagedSpace-cfa640deca460e3aa7362eb9385e98ed

This will delete the workspace from the domain, and users will lose access to their data and notebooks. If you want to back up the data and notebooks first, click Cancel.

Do you want to delete this space?

Yes, delete space

1

To confirm deletion, enter *delete* in the field.

delete

2

Cancel

Delete space

3

- Navigate to the **User Profiles** section and delete all the users.

Amazon SageMaker

Getting started

Applications and IDEs

Studio
Canvas
RStudio
TensorBoard
Profiler
Notebooks

Admin configurations

Domains
Role manager
Images
Lifecycle configurations

SageMaker dashboard

Search

JumpStart

Foundation models

Amazon SageMaker > Domains > Domain: QuickSetupDomain-20240817T120701

QuickSetupDomain-20240817T120701

Domain details

Configure and manage the domain.

Domain settings

User profiles

Space management

App Configurations

Environment

User profiles

Info

A user profile represents a single user within a domain. It is the main way to reference a user for the purposes of sharing, reporting, and other user

Search users

Name	Modified on	Created on
default-20240817T120701		

ult-20240817T120701

Launch **Delete**

Edit

72807:role/service-role/AmazonSageMaker-T120706

Status
✔ InService

ID
d-8uetiqxq3vrd

GMT+0530 (India Standard Time)

Created Action

No apps

Delete user

Do you want to delete this user?

Yes, delete user

To confirm deletion, type *delete* in the field.

delete

Cancel **Delete**

- Lastly, to **delete the domain**, go to the **"Domain Settings,"** scroll down, and click on **"Delete Domain"** to remove it.

Amazon SageMaker > Domains > Domain: QuickSetupDomain-20240817T120701

QuickSetupDomain-20240817T120701

Domain details

Configure and manage the domain.

Domain settings

User profiles

Space management

App Configurations

Environment

User profiles [Info](#)

A user profile represents a single user within a domain. It is the main way to reference a user for the purposes of sharing, reporting, and other user-or

Name



Modified on



Created on

No users

To add a user, choose Add user and

Storage configurations

Encryption key

No custom encryption

Defa

5

Maximum space size

100

Delete domain

To delete a domain, the domain cannot contain any user profiles or spaces. To delete a user profile or space, the profile or space cannot contain any non-failed apps.

Delete domain

Delete Domain



Domain ID

d-8uetiqxq3vrd

This will delete the domain and access to all your notebooks and data.

Do you want to delete the domain?

Yes, delete my Domain

To confirm deletion, type *delete* in the field.

delete

Cancel

Delete

✓ Your domain is pending deletion and this may take a few minutes.

Note: Wait for 2-3 minutes and refresh the page to confirm the deletion.

11 TROUBLESHOOTING

Here, we will add some of the common errors encountered while performing the Lab activity.

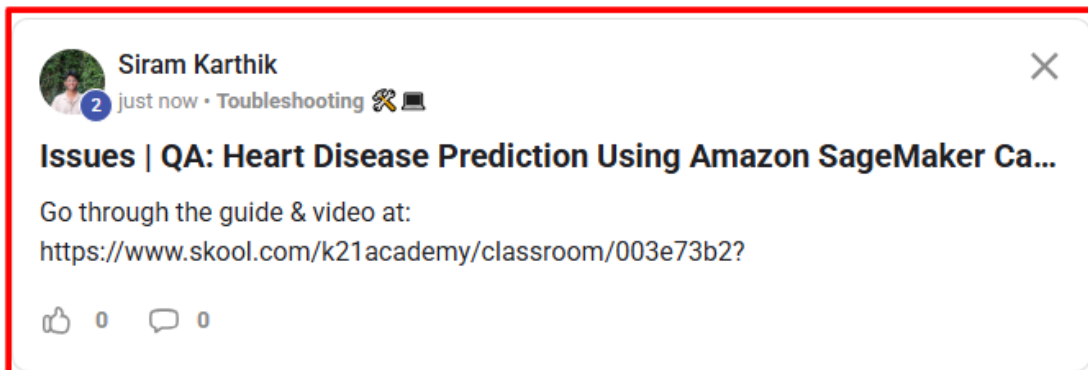
Note: If you encounter any issues in performing this lab, then add your issue in the thread below this lab (linked to this thread in the community):

<https://www.skool.com/k21academy/issues-qa-heart-disease-prediction-using-amazon-sagemaker-canvas-post-here>

Heart Disease Prediction using SageMaker Canvas



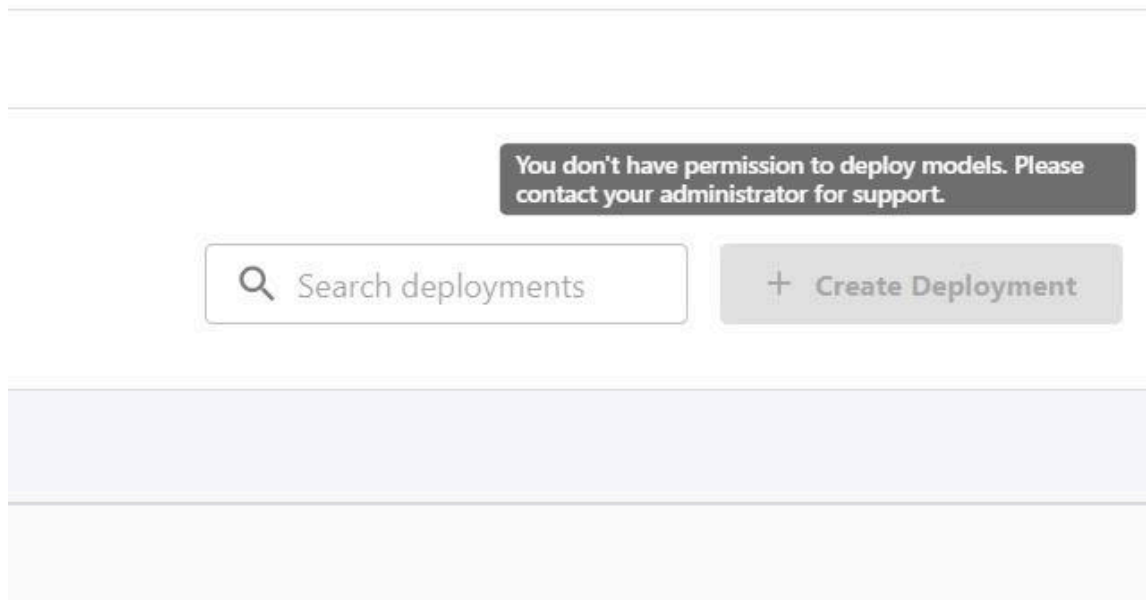
For a detailed, step-by-step project guide, please click here: >>> [Heart Disease Prediction Using Amazon SageMaker Canvas: A No-Code Approach](#)



11.1 You don't have permission to deploy models. Please contact your administrator for support.

11.1.1 ISSUE:

Encountering an error while deploying the machine learning model using SageMaker Canvas.

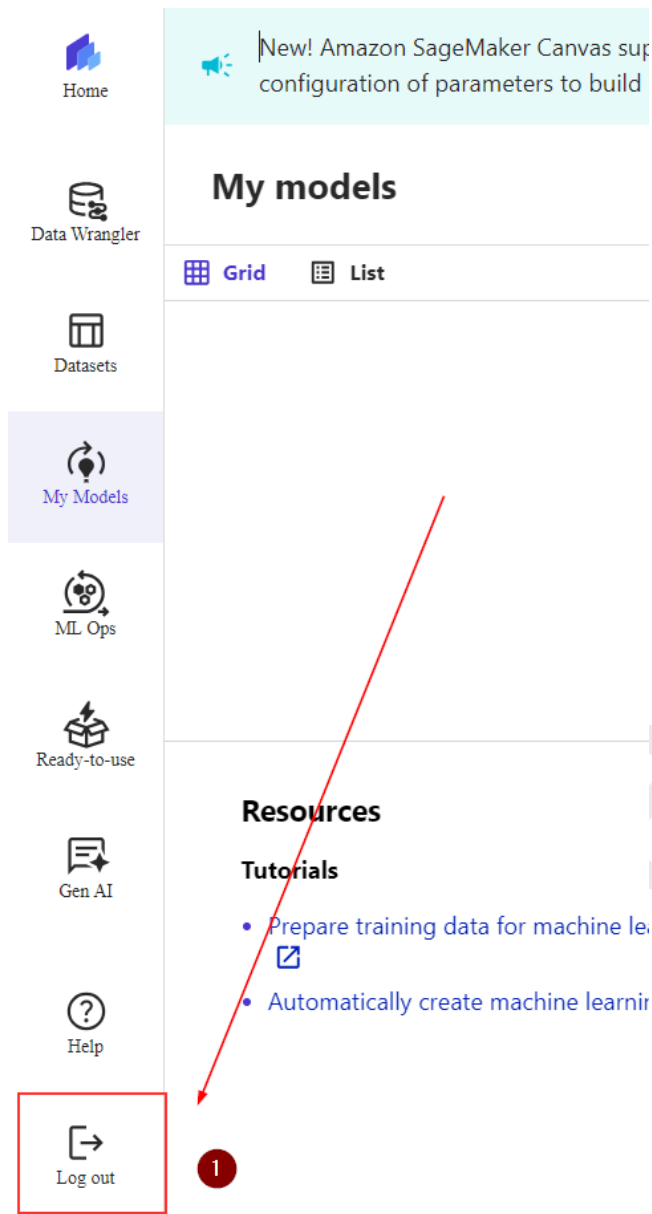


11.1.2 REASON:

Direct deployment of ML models on Canvas is not allowed by default, as the setting is disabled. To resolve this, the necessary permissions must be enabled in the SageMaker Domain.

11.1.3 Fix:

1. Firstly, log out of the SageMaker Canvas.



2. Go to the SageMaker Domain and select **App Configuration**

QuickSetupDomain-20250625T111657

Domain details

Configure and manage the domain.

[Domain settings](#)

[User profiles](#)

[Space management](#)

[App Configurations](#)

[Environment](#)

[Resources](#)

General settings [Info](#)

Name

QuickSetupDomain-20250625T111657

Status

Ready

Domain ID

d-xylutsjovnqf

Created

Wed Jun 25 2025 11:16:40 GMT+0530 (India Standard Time)

Last modified

Wed Jun 25 2025 11:18:38 GMT+0530 (India Standard Time)

VPC

vpc-003b700a3c93a8676

3. Scroll down and navigate to the **Canvas** Configuration section.

4. You will see that the **direct deployment of Canvas models is disabled**. Click the **Edit** option to enable it.


Canvas
1
[Edit](#)

Generate accurate machine learning predictions - no code required.

Canvas configuration

Canvas storage configuration

Amazon S3 artifacts location	Encryption key
-	-

Amazon Q Developer

Enable Amazon Q Developer in SageMaker Canvas for natural language ML	Enable Amazon Q Developer chat for general AWS questions
Disabled	Disabled

Canvas large data processing configuration

Canvas EMR role

arn:aws:iam::442042521384:role/service-role/AmazonSageMakerCanvasEMRSExecutionAccess-20250625T111657

ML Ops configuration

Enable direct deployment of Canvas models	Enable Model Registry registration permissions for all users
Disabled	Disabled

5. **Enable the Direct Deployment of Canvas Models** option under MLOps Configuration, then click on **Submit**.

ML Ops configuration

☒ Enable direct deployment of Canvas models

1

Deploy models to SageMaker endpoints to streamline the process of moving models to production, offering a seamless, scalable, and secure method to make predictions available to end users or applications. If this feature is disabled, you will need to enable direct deployment permissions for each user profile individually. [Learn more](#)

☒ Enable Model Registry registration permissions for all users

Register models to the Model Registry in SageMaker Studio to efficiently manage, version, and deploy models, facilitating collaboration, tracking, and automation throughout the model lifecycle. If this feature is disabled, you will need to enable Model Registry registration permissions for each user profile individually. [Learn more](#)

6. Now, launch SageMaker Canvas, and you will see the deployment option available.

12 SUMMARY

In this guide, we covered:

- Setting up a domain to begin using Amazon SageMaker
- Launching Amazon SageMaker Canvas
- Preparing data with **SageMaker Data Wrangler**
- Building a machine learning model using **Amazon SageMaker Canvas**
- Deploying and testing the model